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BATTERY

Abstract

A battery includes first and second battery cells, a packaging film and a separator. A first sealing edge where tabs of the first and second battery cells are located is formed by the first and second film housings and the separator. A thickness of a hot melt adhesive on the first sealing edge at a side, close to the separator, of a tab of each battery cell of at least one of the first and second battery cells is greater than a thickness of a hot melt adhesive on the first sealing edge at a side, away from the separator, of the tab of the battery cell, which ensures that a certain distance is maintained between a packaging film and a non-separator side of a tab after sealing, so as to reduce a risk of contact short circuits at the tab, thereby improving a safety of a battery.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2024/071816, filed on Jan. 11, 2024, which claims priority to Chinese Patent Application No. 202320248646.6, filed on Feb. 16, 2023. Both of the aforementioned applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] The present disclosure relates to the field of battery technologies, and in particular, to a battery.

BACKGROUND

[0003] A method for improving a working voltage of lithium-ion batteries generally includes: connecting a plurality of identical battery cells in series. In the related art, a separator is usually used to separate and package two identical battery cells that are disposed in parallel. However, since packaging is performed after the separator is added, structures on two sides of a tab become asymmetric after the packaging, such that a non-separator side of the tab is closer to a packaging film (for example, an aluminum laminate film). This makes it impossible to ensure that a safe distance is maintained between the aluminum laminate film and the non-separator side of the tab, thereby resulting in a high risk of contact short circuits at the tab and ultimately resulting in a low safety of a battery.

SUMMARY

[0004] Embodiments of the present disclosure provide a battery, to solve a problem in that, it is impossible to ensure a safe distance between a packaging film (for example, an aluminum laminate film) and a non-separator side of a tab, thereby resulting in a high risk of contact short circuits at the tab and ultimately resulting in a low safety of a battery.

[0005] In a first aspect, embodiments of the present disclosure provide a battery, including a first battery cell, a second battery cell, a packaging film and a separator; the packaging film includes a first film housing and a second film housing, the separator is disposed between the first film housing and the second film housing, the first film housing and the second film housing are separated by the separator, and a first accommodating cavity and a second accommodating cavity that are arranged in parallel are formed by the separator; the first battery cell is accommodated within the first accommodating cavity, and the second battery cell is accommodated within the second accommodating cavity; a plurality of sealing edges are formed by the first film housing, the separator and the second film housing, and the plurality of sealing edges include a first sealing edge where a tab of the first battery cell and a tab of the second battery cell are located; and a thickness of a hot melt adhesive on the first sealing edge at a side, close to the separator, of a tab of each battery cell of at least one of the first battery cell and the second battery cell is greater than a thickness of a hot melt adhesive on the first sealing edge at a side, away from the separator, of the tab of the battery cell.

[0006] Optionally, a first side of the tab of the first battery cell is a side, close to the separator, of the tab of the first battery cell, a second side of the tab of the first battery cell is a side, away from the separator, of the tab of the first battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell is greater than the thickness of

the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell.

[0007] Optionally, the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell includes a first tab adhesive layer, the separator and a first resin layer that are stacked sequentially, and the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell includes a second tab adhesive layer and a second resin layer that are stacked sequentially, where the first tab adhesive layer and the second tab adhesive layer are located on both sides of the tab of the first battery cell, and the first resin layer and the second resin layer are film layers, close to the tab of the first battery cell, of the packaging film.

[0008] Optionally, a ratio of the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell to the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell is between 1.01 and 5.

[0009] Optionally, a difference between the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell and the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell ranges from 10 μm to 500 μm .

[0010] Optionally, a first side of the tab of the second battery cell is a side, away from the separator, of the tab of the second battery cell, a second side of the tab of the second battery cell is a side, close to the separator, of the tab of the second battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell is greater than the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell.

[0011] Optionally, the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell includes a third tab adhesive layer and a first resin layer that are stacked sequentially, and the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell includes a fourth tab adhesive layer, the separator and a second resin layer that are stacked sequentially, where the third tab adhesive layer and the fourth tab adhesive layer are located on both sides of the tab of the second battery cell, and the first resin layer and the second resin layer are film layers, close to the tab of the second battery cell, of the packaging film.

[0012] Optionally, a ratio of the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell to the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell is between 1.01 and 5.

[0013] Optionally, a difference between the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell and the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell ranges from 10 μm to 500 μm .

[0014] Optionally, a first side of the tab of the first battery cell is a side, close to the separator, of the tab of the first battery cell, a first side of the tab of the second battery cell is a side, away from the separator, of the tab of the second battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell is greater than the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell.

[0015] Optionally, a second side of the tab of the first battery cell is a side, away from the separator, of the tab of the first battery cell, a second side of the tab of the second battery cell is a side, close to the separator, of the tab of the second battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell is less than the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell.

[0016] Optionally, a first side of the tab of the first battery cell is a side, close to the separator, of the tab of the first battery cell, a second side of the tab of the second battery cell is a side, close to the separator, of the tab of the second battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell is equal to the thickness of the

hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell.

[0017] Optionally, a second side of the tab of the first battery cell is a side, away from the separator, of the tab of the first battery cell, a first side of the tab of the second battery cell is a side, away from the separator, of the tab of the second battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell is equal to the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell.

[0018] Optionally, the plurality of sealing edges further includes a second sealing edge, a third sealing edge and a fourth sealing edge, the second sealing edge and the first sealing edge are disposed opposite to each other, and the third sealing edge and the fourth sealing edge are disposed opposite to each other; and compression ratios of the first sealing edge, the second sealing edge, the third sealing edge and the fourth sealing edge are all between 3% and 97%.

[0019] Optionally, the separator includes a first region recessed toward a bottom of the first film housing, and a distance between the first region and the second film housing is equal to an accommodating depth of the first film housing.

[0020] Optionally, the accommodating depth of the first film housing is greater than an accommodating depth of the second film housing.

[0021] Optionally, the separator further includes a second region recessed toward a bottom of the second film housing, and a distance between the second region and the bottom of the first film housing is equal to the distance between the first region and the second film housing.

[0022] In the embodiments of the present disclosure, the battery includes the first battery cell, the second battery cell, the packaging film and the separator, the packaging film includes the first film housing and the second film housing, the separator is disposed between the first film housing and the second film housing, the first film housing and the second film housing are separated by the separator, and the first accommodating cavity and the second accommodating cavity that are arranged in parallel are formed by the separator; the first battery cell is accommodated within the first accommodating cavity, and the second battery cell is accommodated within the second accommodating cavity; the plurality of sealing edges are formed by the first film housing, the separator and the second film housing, and the plurality of sealing edges include the first sealing edge where the tab of the first battery cell and the tab of the second battery cell are located; and a thickness of a hot melt adhesive on the first sealing edge at a side, close to the separator, of a tab of each battery cell of at least one of the first battery cell and the second battery cell is greater than a thickness of a hot melt adhesive on the first sealing edge at a side, away from the separator, of the tab of the battery cell. In this way, this ensures that a certain distance is maintained between a packaging film (for example, an aluminum laminate film) and a non-separator side of a tab after sealing, so as to reduce a risk of contact short circuits at the tab, thereby improving a safety of a battery.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] In order to more clearly illustrate technical solutions of the present disclosure, the following briefly describes the accompanying drawings required in the description of embodiments of the present disclosure. Obviously, the accompanying drawings in the following description are only some embodiments of the present disclosure, and as for a person of ordinary skill in the art, other accompanying drawings may also be obtained according to these accompanying drawings without creative efforts.

[0024] FIG. 1 is a schematic structural diagram of a battery before packaging according to an embodiment of the present disclosure.

[0025] FIG. 2 is a cross-sectional schematic diagram of the battery of FIG. 1.

[0026] FIG. 3 is a schematic structural diagram of a battery after packaging according to an embodiment of the present disclosure.

[0027] FIG. 4 is a schematic structural diagram of a battery according to another embodiment of the present disclosure.

[0028] FIG. 5 is a schematic diagram of a sealing head structure adapted to a tab structure during sealing according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0029] The following clearly and completely describes the technical solutions in the embodiments of the present disclosure with reference to the accompanying drawings in the embodiments of the present disclosure. Obviously, the described embodiments are a part rather than all of the embodiments of the present disclosure. All other embodiments obtained by a person of ordinary skill in the art based on the embodiments of the present disclosure without creative efforts shall fall within the protection scope of the present disclosure.

[0030] Unless otherwise defined, technical terms or scientific terms used in the present disclosure should have the common meanings understood by one of ordinary skill in the art to which the present disclosure belongs. The terms “first”, “second”, and similar terms used in the present disclosure do not indicate any order, quantity or importance, but are merely used to distinguish different components. Likewise, the similar terms such as “a” or “an” do not denote a quantity limitation, but indicate that there is at least one. The similar terms such as “connect” or “join” are not limited to physical or mechanical connections, but may include electrical connections, whether direct or indirect. The terms “up”, “down”, “left”, “right”, etc. are only used to represent a relative positional relationship, and when an absolute position of a described object changes, the relative positional relationship also changes correspondingly.

[0031] Taking two identical battery cells that are disposed in parallel as an example, a Z-shaped separator is disposed between two packaging housings of equal size, the separator is then in sealed connection to the upper packaging housing and the lower packaging housing, respectively, to realize packaging, and finally the two battery cells are disposed within an accommodating cavity formed by the separator and the upper packaging housing, and an accommodating cavity formed by the separator and the lower packaging housing, respectively. However, since the packaging is performed after the separator is added, structures on two sides of a tab become asymmetric after the packaging, such that a non-separator side of the tab is closer to a packaging film (for example, an aluminum laminate film). This makes it impossible to ensure that a safe distance is maintained between the aluminum laminate film and the non-separator side of the tab, thereby resulting in a high risk of contact short circuits at the tab and ultimately resulting in a low safety of a battery.

[0032] Referring to FIG. 1 to FIG. 3, embodiments of the present disclosure provide a battery, including a first battery cell **10**, a second battery cell **20**, a packaging film and a separator **30**; the packaging film includes a first film housing **40** and a second film housing **50**, the separator **30** is disposed between the first film housing **40** and the second film housing **50**, the first film housing **40** and the second film housing **50** are separated by the separator **30**, and a first accommodating cavity and a second accommodating cavity that are arranged in parallel are formed by the separator **30**; the first battery cell **10** is accommodated within the first accommodating cavity, and the second battery cell **20** is accommodated within the second accommodating cavity; a plurality of sealing edges **60** are formed by the first film housing **40**, the separator **30** and the second film housing **50**, and the plurality of sealing edges **60** include a first sealing edge **61** where a tab **11** of the first battery cell **10** and a tab **21** of the second battery cell **20** are located; and a thickness of a hot melt adhesive on the first sealing edge **61** at a side, close to the separator **30**, of a tab of each battery cell of at least one of the first battery cell **10** and the second battery cell **20** is greater than a thickness of a hot melt adhesive on the first sealing edge **61** at a side, away from the separator **30**, of the tab of the battery cell.

[0033] In some embodiments, a first side of the tab **11** of the first battery cell **10** is a side, close to the separator **30**, of the tab **11** of the first battery cell **10**, a second side of the tab **11** of the first battery cell **10** is a side, away from the separator **30**, of the tab **11** of the first battery cell **10**, and the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **11** of the first battery cell **10** is greater than the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **11** of the first battery cell **10**.

[0034] In some other embodiments, a first side of the tab **21** of the second battery cell **20** is a side, away from the separator **30**, of the tab **21** of the second battery cell **20**, a second side of the tab **21** of the second battery cell **20** is a side, close to the separator **30**, of the tab **21** of the second battery cell **20**, and the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **21** of the second battery cell **20** is greater than the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **21** of the second battery cell **20**.

[0035] It should be understood that the separator **30** may be a Z-shaped structure, and the Z-shaped separator **30**, the first film housing **40** and the second film housing **50** form two accommodating cavities that are arranged in parallel. The tab **11** of the first battery cell **10** disposed within the first accommodating cavity and the tab **21** of the second battery cell **20** disposed within the second accommodating cavity extend out of the first accommodating cavity and the second accommodating cavity respectively through different regions of the first sealing edge **61**. It should be noted that the Z-shaped separator **30** also participates in packaging. During packaging, since the Z-shaped separator **30** may separate the two accommodating cavities, the Z-shaped separator **30** may also play a role in limiting the first battery cell **10** and the second battery cell **20**, thereby further improving overall packaging strength.

[0036] It should be understood that the first film housing **40** and the second film housing **50** may both be formed of aluminum laminate films. For example, as shown in FIG. 4, the aluminum laminate film may be composed of a PP (Polypropylene) layer, an aluminum layer and a nylon layer that are disposed sequentially, and the PP layer is located on a side close to the tab.

[0037] The separator **30** is the Z-shaped structure, such that the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **11** of the first battery cell **10** includes a thickness of a tab adhesive (i.e., a first tab adhesive layer) adhered to the first side of the tab **11** of the first battery cell **10**, a thickness of the separator **30** and a thickness of a PP layer (i.e., a first resin layer) of the second film housing **50**, and the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **11** of the first battery cell **10** includes a thickness of a tab adhesive (i.e., a second tab adhesive layer) adhered to the second side of the tab **11** of the first battery cell **10** and a thickness of a PP layer (i.e., a second resin layer) of the first film housing **40**.

[0038] The first tab adhesive layer and the second tab adhesive layer are located on both sides of the tab **11** of the first battery cell **10**, and the first resin layer and the second resin layer are film layers, close to the tab **11** of the first battery cell **10**, of the packaging film. For example, the first resin layer is the film layer, close to the tab **11** of the first battery cell **10**, of the second film housing **50**, and the second resin layer is the film layer, close to the tab **11** of the first battery cell **10**, of the first film housing **40**.

[0039] Similarly, the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **21** of the second battery cell **20** includes a thickness of a tab adhesive (i.e., a third tab adhesive layer) adhered to the first side of the tab **21** of the second battery cell **20** and a thickness of a PP layer (i.e., a first resin layer) of the second film housing **50**, and the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **21** of the second battery cell **20** includes a thickness of a tab adhesive (i.e., a fourth tab adhesive layer) adhered to the second side of the tab **21** of the second battery cell **20**, a thickness of the separator **30**, and a thickness of a PP layer (i.e., a second resin layer) of the first film housing **40**.

[0040] The third tab adhesive layer and the fourth tab adhesive layer are located on both sides of the tab **21** of the second battery cell **20**, the first resin layer and the second resin layer are film

layers, close to the tab **21** of the second battery cell **20**, of the packaging film. For example, the first resin layer is the film layer, close to the tab **21** of the second battery cell **20**, of the second film housing **50**, and the second resin layer is the film layer, close to the tab **21** of the second battery cell **20**, of the first film housing **40**.

[0041] It should be noted that the separator **30** may be divided into a single-layer structure or a multi-layer composite structure, so as to meet different requirements of different battery structures. For example, a material of the separator **30** with the single-layer structure includes a polymer material or a carbon material; and a material of the separator **30** with the multi-layer composite structure includes a metal material of an intermediate layer, a polymer material or a carbon material of an outer layer, and a polymer material of a bonding layer located between the intermediate layer and the outer layer. The intermediate layer is used for separation and support, the outer layer is used for insulation, and the bonding layer is used for bonding the intermediate layer to the outer layer.

[0042] The metal material may include at least one of Ni, Ti, Cu, Ag, Au, Pt, Fe, Co, Cr, W, Mo, Al, Mg, K, Na, Ca, Sr, Ba, Si, Ge, Sb, Pb, In, Zn, Stainless Steel (SUS), and their compounds or alloys, etc.; the polymer material may include at least one of polyethylene terephthalate, polybutylene terephthalate, polyethylene naphthalate, polyether ether ketone, polyimide, polyamide, polyethylene glycol, polyamide-imide, polycarbonate, cyclic olefin copolymer, polyphenylene sulfide, polyvinyl acetate, polytetrafluoroethylene, polynaphthylene methylene, polyvinylidene fluoride, polyethylene furanoate, polypropylene carbonate, poly(vinylidene fluoride-co-hexafluoropropylene), poly(vinylidene fluoride-co-chlorotrifluoroethylene), silicone, vinylon, polypropylene, maleic anhydride-modified polypropylene, polyethylene, ethylene and its copolymers, polyvinyl chloride, polystyrene, polyether nitrile, polyurethane, polyphenylene oxide, polyester, polysulfone, amorphous alpha-olefin copolymer and their derivatives; and the carbon material may include at least one of carbon felt, carbon film, carbon black, acetylene black, fullerene, conductive graphite film or graphene film.

[0043] In the embodiments of the present disclosure, the battery includes the first battery cell **10**, the second battery cell **20**, the packaging film and the separator **30**. The packaging film includes the first film housing **40** and the second film housing **50**, the separator **30** is disposed between the first film housing **40** and the second film housing **50**, the first film housing **40** and the second film housing **50** are separated by the separator **30**, and the first accommodating cavity and the second accommodating cavity that are arranged in parallel are formed by the separator **30**. The first battery cell **10** is accommodated within the first accommodating cavity, and the second battery cell **20** is accommodated within the second accommodating cavity. The plurality of sealing edges **60** are formed by the first film housing **40**, the separator **30** and the second film housing **50**, and the plurality of sealing edges **60** include the first sealing edge **61** where the tab **11** of the first battery cell **10** and the tab **21** of the second battery cell **20** are located. The thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **11** of the first battery cell **10** is greater than the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **11** of the first battery cell **10**, where the first side of the tab **11** of the first battery cell **10** is the side, close to the separator **30**, of the tab **11** of the first battery cell **10**, and the second side of the tab **11** of the first battery cell **10** is the side, away from the separator **30**, of the tab **11** of the first battery cell **10**. The thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **21** of the second battery cell **20** is less than the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **21** of the second battery cell **20**, where the first side of the tab **21** of the second battery cell **20** is the side, away from the separator **30**, of the tab **21** of the second battery cell **20**, and the second side of the tab **21** of the second battery cell **20** is the side, close to the separator **30**, of the tab **21** of the second battery cell **20**. In this way, this ensures that a certain distance is maintained between an aluminum laminate film and a non-separator side of a tab after sealing, so as to reduce a risk of contact short circuits at the tab, thereby improving a

safety of a battery.

[0044] In an optional embodiment, the hot melt adhesive on the first sealing edge **61** at the first side of the tab **11** of the first battery cell **10** includes a tab adhesive layer, the separator, and a resin layer that are stacked sequentially, and the hot melt adhesive on the first sealing edge **61** at the first side of the tab **21** of the second battery cell **20** includes a tab adhesive layer and a resin layer that are stacked sequentially; and the hot melt adhesive on the first sealing edge **61** at the second side of the tab **11** of the first battery cell **10** includes a tab adhesive layer and a resin layer that are stacked sequentially, and the hot melt adhesive on the first sealing edge **61** at the second side of the tab **21** of the second battery cell **20** includes a tab adhesive layer, the separator **30** and a resin layer that are stacked sequentially.

[0045] In an optional embodiment, a ratio of the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **11** of the first battery cell **10** to the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **11** of the first battery cell **10** is between 1.01 and 5.

[0046] In an optional embodiment, a ratio of the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **21** of the second battery cell **20** to the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **21** of the second battery cell **20** is between 1.01 and 5.

[0047] In an optional implementation, the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **11** of the first battery cell **10** is greater than the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **21** of the second battery cell **20**, and the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **11** of the first battery cell **10** is less than the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **21** of the second battery cell **20**.

[0048] In another optional implementation, the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **11** of the first battery cell **10** is equal to the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **21** of the second battery cell **20**.

[0049] In another optional implementation, the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **11** of the first battery cell **10** is equal to the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **21** of the second battery cell **20**.

[0050] In an optional implementation, a difference between the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **11** of the first battery cell **10** and the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **11** of the first battery cell **10** ranges from 10 μm to 500 μm ; and a difference between the thickness of the hot melt adhesive on the first sealing edge **61** at the first side of the tab **21** of the second battery cell **20** and the thickness of the hot melt adhesive on the first sealing edge **61** at the second side of the tab **21** of the second battery cell **20** ranges from 10 μm to 500 μm .

[0051] Optionally, the plurality of sealing edges **60** further includes a second sealing edge **62**, a third sealing edge **63** and a fourth sealing edge **64**, the second sealing edge **62** and the first sealing edge **61** are disposed opposite to each other, and the third sealing edge **63** and the fourth sealing edge **64** are disposed opposite to each other; and compression ratios of the second sealing edge **62**, the third sealing edge **63** and the fourth sealing edge **64** are all between 3% and 97%.

[0052] It should be understood that the compression ratio is a ratio of a thickness of a sealing edge **60** of the battery before packaging to a thickness of the sealing edge **60** of the battery after packaging.

[0053] When the battery is of a square structure, i.e., the first film housing **40**, the separator **30** and the second film housing **50** form four sealing edges **60**, the first sealing edge **61** is a sealing edge where the tabs of the two battery cells are located. The compression ratios of the other three sealing

edges **60** are all set between 3% and 97%, so that a thickness difference between the plurality of sealing edges **60** after packaging is reduced, thereby improving packaging strength.

[0054] In an optional embodiment, the separator **30** includes a first region **31** recessed toward a bottom of the first film housing **40**.

[0055] It should be understood that, since the separator **30**, the first film housing **40** and the second film housing **50** form the two accommodating cavities that are arranged in parallel, and the plurality of sealing edges **60** formed by the first film housing **40**, the separator **30** and the second film housing **50** are located on a same horizontal plane, when the first region **31** of the separator **30** recesses toward the bottom of the first film housing **40**, the second accommodating cavity is formed by the second film housing **50** and the first region **31** of the separator **30**. In this case, the second battery cell **20** is located on a side of the first region **31** of the separator **30**, and the first battery cell **10** is located on another side of the separator **30**. An accommodating depth of the first film housing **40** is equal to a sum of a thickness of the first battery cell **10** and a thickness of the separator **30**; or an accommodating depth of the first film housing **40** is equal to a sum of a thickness of the second battery cell **20** and a thickness of the separator **30**, and an accommodating depth of the second film housing **50** is equal to 0.

[0056] In another optional embodiment, the separator **30** further includes a second region **32** recessed toward a bottom of the second film housing **50**, and a distance between the second region **32** and the bottom of the first film housing **40** is equal to a distance between the first region **31** and the bottom of the second film housing **50**.

[0057] In this embodiment, the separator **30** is set to further include the second region **32** recessed toward the bottom of the second film housing **50**, the distance between the second region **32** and the bottom of the first film housing **40** is also set to be the same as the distance between the first region **31** and the bottom of the second film housing **50**, and the thickness of the first battery cell **10** disposed on a side of the second region **32** is also set to be the same as the thickness of the second battery cell **20** disposed on a side of the first region **31**, thereby improving voltage stability of a battery.

[0058] Optionally, the accommodating depth of the first film housing **40** is greater than the accommodating depth of the second film housing **50**.

[0059] It should be understood that, since a sum of the accommodating depth of the first film housing **40** and the accommodating depth of the second film housing **50** is equal to a sum of the thickness of the first battery cell **10**, the thickness of the second battery cell **20** and the thickness of the separator **30**, the accommodating depth of the first film housing **40** may be set greater than the accommodating depth of the second film housing **50**, that is, the separator **30** is located on the plurality of sealing edges **60**, and a distance between the separator **30** and the bottom of the first film housing **40** is greater than a distance between the separator **30** and the bottom of the second film housing **50**.

[0060] In an optional implementation, the accommodating depth of the second film housing **50** is greater than or equal to 0.

[0061] Since the sum of the accommodating depth of the first film housing **40** and the accommodating depth of the second film housing **50** is equal to the sum of the thickness of the first battery cell **10**, the thickness of the second battery cell **20** and the thickness of the separator **30**, when the accommodating depth of the second film housing **50** is equal to 0, the accommodating depth of the first film housing **40** is the greatest. Correspondingly, a recessed depth of the first region **31** of the separator **30** recessed toward the bottom of the first film housing **40** is the greatest, so that a space of the accommodating cavity formed by the separator **30** and the second film housing **50** may accommodate the second battery cell **20**.

[0062] When the accommodating depth of the second film housing **50** is greater than 0, the accommodating depth of the first film housing **40** is related to the accommodating depth of the second film housing **50**. Correspondingly, a recessed depth of the first region **31** of the separator **30**

recessed toward the bottom of the first film housing **40** is also related to the accommodating depth of the second film housing **50**, which is not limited in this embodiment.

[0063] It should be understood that the accommodating depth mentioned in the present disclosure is specifically the distance D shown in FIG. 3.

[0064] In addition, since the plurality of sealing edges **60** of the battery provided in the embodiments of the present disclosure have different compression ratios, a sealing head structure, configured to adapt to the sealing edges **60** in the embodiments of the present disclosure, may be used during thermal sealing.

[0065] It should be understood that, since thicknesses of melt adhesives on two sides of a tab of a battery cell are not equal after the battery cell is packaged with a separator, and the thickness of the melt adhesive on a separator side is greater than the thickness of the melt adhesive on another side by 10 μm to 500 μm , i.e., since an introduction of the separator **30** brings asymmetry to a packaging structure of the tab of a battery, a multi-tab stepped sealing head, configured to adapt to the sealing edges **60** in the embodiments of the present disclosure, may be used during thermal sealing. For example, as shown in FIG. 5, a sum of a limit of an upper sealing head and a limit of a lower sealing head is equal to a sum of a value b of a limit on a separator side and a value c of a limit on a non-separator side, and $b > c$.

[0066] As shown in FIG. 4, the introduction of the separator **30** brings asymmetry to the packaging structure of the tab of the battery, where the thickness of the melt adhesive on the separator side is greater than the thickness of the melt adhesive on the non-separator side. After sealing, a thickness of the melt adhesive on the separator side is equal to a sum of a thickness of a PP layer of an aluminum laminate film, a thickness of the separator **30** and a thickness of a tab adhesive layer on a tab side, and a thickness of the melt adhesive on the non-separator side is equal to a sum of a thickness of a PP layer of an aluminum laminate film and a thickness of a tab adhesive layer on a tab side. On the separator side of the tab of the battery provided in the present disclosure, a single-layer compression rate of the separator **30** (i.e., the thickness of the melt adhesive) may be in a range of 3% to 97%, and a total thickness of the melt adhesive on the separator side of the tab is A1. A total thickness of the melt adhesive on the non-separator side of the tab is A2. A total compression rate is in a range of 3% to 97%, and a ratio of the thicknesses of the melt adhesives on two sides is $A1/A2$, which is between 1.01 to 5.

[0067] It should be understood that the compression ratio mentioned in the present disclosure refers to a ratio of a thickness after an edge sealing process to a thickness before an edge sealing process.

[0068] The above are merely specific embodiments of the present disclosure, but not intend to limit the protection scope of the present disclosure. Any person skilled in the art can easily conceive of modification or replacement within the technical scope disclosed in the present disclosure, which should all be covered within the protection scope of the present disclosure. Therefore, the protection scope of the present disclosure shall be subject to the protection scope of claims.

Claims

1. A battery, comprising a first battery cell, a second battery cell, a packaging film and a separator, wherein the packaging film comprises a first film housing and a second film housing, the separator is disposed between the first film housing and the second film housing, the first film housing and the second film housing are separated by the separator, and a first accommodating cavity and a second accommodating cavity that are arranged in parallel are formed by the separator; the first battery cell is accommodated within the first accommodating cavity, and the second battery cell is accommodated within the second accommodating cavity; a plurality of sealing edges are formed by the first film housing, the separator and the second film housing, and the plurality of sealing edges comprise a first sealing edge where a tab of the first battery cell and a tab of the second battery cell are located; and a thickness of a hot melt adhesive on the first sealing edge at a side, close to the

separator, of a tab of each battery cell of at least one of the first battery cell and the second battery cell is greater than a thickness of a hot melt adhesive on the first sealing edge at a side, away from the separator, of the tab of the battery cell.

2. The battery according to claim 1, wherein a first side of the tab of the first battery cell is a side, close to the separator, of the tab of the first battery cell, a second side of the tab of the first battery cell is a side, away from the separator, of the tab of the first battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell is greater than the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell.

3. The battery according to claim 2, wherein the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell comprises a first tab adhesive layer, the separator and a first resin layer that are stacked sequentially, and the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell comprises a second tab adhesive layer and a second resin layer that are stacked sequentially, the first tab adhesive layer and the second tab adhesive layer are located on both sides of the tab of the first battery cell, and the first resin layer and the second resin layer are film layers, close to the tab of the first battery cell, of the packaging film.

4. The battery according to claim 2, wherein a ratio of the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell to the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell is between 1.01 and 5.

5. The battery according to claim 2, wherein a difference between the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell and the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell ranges from 10 μm to 500 μm .

6. The battery according to claim 1, wherein a first side of the tab of the second battery cell is a side, away from the separator, of the tab of the second battery cell, a second side of the tab of the second battery cell is a side, close to the separator, of the tab of the second battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell is greater than the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell.

7. The battery according to claim 6, wherein the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell comprises a third tab adhesive layer and a first resin layer that are stacked sequentially, and the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell comprises a fourth tab adhesive layer, the separator and a second resin layer that are stacked sequentially, the third tab adhesive layer and the fourth tab adhesive layer are located on both sides of the tab of the second battery cell, and the first resin layer and the second resin layer are film layers, close to the tab of the second battery cell, of the packaging film.

8. The battery according to claim 6, wherein a ratio of the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell to the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell is between 1.01 and 5.

9. The battery according to claim 6, wherein a difference between the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell and the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell ranges from 10 μm to 500 μm .

10. The battery according to claim 1, wherein a first side of the tab of the first battery cell is a side, close to the separator, of the tab of the first battery cell, a first side of the tab of the second battery cell is a side, away from the separator, of the tab of the second battery cell, and the thickness of the

hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell is greater than the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell.

11. The battery according to claim 1, wherein a second side of the tab of the first battery cell is a side, away from the separator, of the tab of the first battery cell, a second side of the tab of the second battery cell is a side, close to the separator, of the tab of the second battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell is less than the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell.

12. The battery according to claim 1, wherein a first side of the tab of the first battery cell is a side, close to the separator, of the tab of the first battery cell, a second side of the tab of the second battery cell is a side, close to the separator, of the tab of the second battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the first battery cell is equal to the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the second battery cell.

13. The battery according to claim 1, wherein a second side of the tab of the first battery cell is a side, away from the separator, of the tab of the first battery cell, a first side of the tab of the second battery cell is a side, away from the separator, of the tab of the second battery cell, and the thickness of the hot melt adhesive on the first sealing edge at the second side of the tab of the first battery cell is equal to the thickness of the hot melt adhesive on the first sealing edge at the first side of the tab of the second battery cell.

14. The battery according to claim 1, wherein the plurality of sealing edges further comprises a second sealing edge, a third sealing edge and a fourth sealing edge, the second sealing edge and the first sealing edge are disposed opposite to each other, and the third sealing edge and the fourth sealing edge are disposed opposite to each other; and compression ratios of the first sealing edge, the second sealing edge, the third sealing edge and the fourth sealing edge are all between 3% and 97%.

15. The battery according to claim 1, wherein the separator comprises a first region recessed toward a bottom of the first film housing, and a distance between the first region and the second film housing is equal to an accommodating depth of the first film housing.

16. The battery according to claim 15, wherein the accommodating depth of the first film housing is greater than an accommodating depth of the second film housing.

17. The battery according to claim 15, wherein the separator further comprises a second region recessed toward a bottom of the second film housing, and a distance between the second region and the bottom of the first film housing is equal to the distance between the first region and the second film housing.
